Part A: Quant Assignment Solution

25B0315

Part 1: Fixed Strategy finding Expected Value

The game uses a fair 100-sided die. The strategy is simple: stop and collect \$X if the roll is above 50, otherwise re-roll. Each re-roll carries a 3% crash risk (payout \$0) and 97% survival chance.

Since surviving a re-roll brings you back to the same game state, we can write a recursive equation for the expected value E_f.

Step 1- The stop case. Rolling above 50 happens with probability $\frac{50}{100} = 0.5$. The conditional expected payout is the average of integers from 51 to 100:

$$E[X \mid X > 50] = \frac{51 + 100}{2} = 75.5$$

Step 2 - The re-roll case. Rolling 50 or below also happens with probability 0.5. When we re-roll, there is a crash risk, so:

$$E[\text{re-roll}] = 0.03 \times 0 + 0.97 \times E_f = 0.97\,E_f$$

Step 3 - after Putting it together.

$$E_f = 0.5 \times 75.5 + 0.5 \times 0.97\,E_f$$

$$E_f = 37.75 + 0.485\,E_f$$

$$0.515\,E_f = 37.75$$

$$E_f = \frac{37.75}{0.515} = \frac{37750}{515} = \frac{7550}{103} \approx \boxed{\$73.30}$$

Part 2: Optimal Strategy and $E_{\max}$

Now instead of a fixed threshold of 50, we want to find the *best* threshold. let's Call it a the player re-rolls on anything $\leq a$ and stops on anything $> a$. The expected value is then a function of a:

$$E(a) = \left(\frac{100-a}{100}\right)\left(\frac{a+101}{2}\right) + \left(\frac{a}{100}\right)(0.97\,E(a))$$

Isolating $E(a)$:

$$E(a)\left(1 - \frac{0.97a}{100}\right) = \frac{(100-a)(a+101)}{200}$$

After simplifying the numerator $(100-a)(a+101) = 10100 + 100a - 101a - a^2 = 10100 - a - a^2$:

$$E(a) = \frac{10100 - a - a^2}{200 - 1.94a} \tag{1}$$

To find the optimal a, we differentiate with respect to a and set equal to zero. Using the quotient rule with $u = 10100 - a - a^2$ and $v = 200 - 1.94a$:

$$\frac{dE}{da} = \frac{u'v - uv'}{v^2} = 0 \implies u'v - uv' = 0$$

Computing:

$$\begin{aligned}
u' &= -1 - 2a, \quad v' = -1.94 \\
(-1-2a)(200-1.94a) - (10100 - a - a^2)(-1.94) &= 0 \\
-200 + 1.94a - 400a + 3.88a^2 + 19594 - 1.94a - 1.94a^2 &= 0 \\
1.94a^2 - 400a + 19394 &= 0
\end{aligned}$$

Dividing through by 2:

$$0.97a^2 - 200a + 9697 = 0$$

Quadratic formula gives:

$$a = \frac{200 - \sqrt{40000 - 4(0.97)(9697)}}{1.94} = \frac{200 - \sqrt{2425.24}}{1.94} \approx \frac{200 - 48.74}{1.94} \approx 77.97$$

Indifference Point check via Method 2. A second way to see this: at the optimal threshold, the player must be indifferent between stopping at value V and re-rolling. So the value of stopping equals the value of re-rolling:

$$V = 0.97\,E_{\max} \implies E_{\max} = \frac{V}{0.97}$$

Plugging this into the expected value equation and solving the resulting quadratic gives $V \approx 78.46$, which is consistent with $a \approx 77.97$ from above both the methods point to the same integer boundary.

Integer threshold. Since the die only gives integers, we test $a = 78$ (re-roll on 1–78, stop on 79–100):

$$
\begin{aligned}
P(\text{stop}) &= \frac{22}{100} = 0.22, \quad E[\text{roll} \mid \text{stop}] = \frac{79+100}{2} = 89.5 \\
E_{\max} &= 0.22 \times 89.5 + 0.78 \times 0.97 \times E_{\max} \\
E_{\max} &= 19.69 + 0.7566\, E_{\max} \\
0.2434\, E_{\max} &= 19.69 \\
E_{\max} &= \frac{19.69}{0.2434} \approx \boxed{\$80.90}
\end{aligned}
$$

Empirical confirmation. Evaluating equation (1) across nearby integer thresholds confirms $a = 78$ is the peak:

Threshold a	$E(a)$
75	\$80.7339
76	\$80.8219
77	\$80.8771
78	**\$80.8956**
79	\$80.8729
80	\$80.8036
81	\$80.6813

Table 1: Expected value around the optimal integer threshold.

The function increases up to $a = 78$ and falls after as the 3% crash risk starts outweighing the benefit of chasing higher numbers beyond this point. The optimal strategy is therefore: **re-roll on 1–78, stop on 79–100**, yielding $E_{\max} \approx \$80.90$.

Part 3: Crash Insurance — Should You Buy It?

You have just rolled a 75. A bookie offers crash insurance for the next roll only, at a flat fee of \$2. If you pay and a crash happens, you still get your \$75. Three options are on the table.

Risk-Neutral Expected Value

We already know $E_{\max} = \$80.90$ is the value of a fresh roll under optimal play.

Choice A — Stop now. Just take the \$75. EV = \$75.00.

Choice B — Re-roll without insurance.

$$EV_B = 0.03 \times 0 + 0.97 \times 80.90 = \$78.47$$

Choice C — Re-roll with insurance (\$2 fee).

$$EV_C = 0.03 \times 75 + 0.97 \times 80.90 - 2.00 = 2.25 + 78.47 - 2.00 = \$78.72$$

So the ranking is:

$$\text{Choice C (\$78.72)} > \text{Choice B (\$78.47)} > \text{Choice A (\$75.00)}$$

Why is insurance a buy? The fair value of the insurance is simply:

$$0.03 \times \$75 = \$2.25$$

The bookie is charging \$2.00 for something worth \$2.25 and that is a \$0.25 edge, which exactly equals $EV_C - EV_B$. So buying the insurance is a correct decision purely on arbitrage grounds.

Risk-Averse Approach (Expected Utility Theory)

A purely EV-maximizing framework ignores the fact that most people *hate* losing everything. To model this, we use a logarithmic utility function:

$$U(x) = \ln(1 + x)$$

The +1 shift ensures $U(0) = 0$ (wipeout is not infinitely bad, just zero utility).

Under this framework, the optimal stopping threshold shifts from $a = 78$ down to $a = 57$ because the logarithm penalises the 3% ruin scenario much more heavily.

Deriving $EU_{\max}$. The recursive structure is exactly the same as Part 1, just with utility values instead of dollar values. For threshold a, stopping on roll $x > a$ gives utility $\ln(1 + x)$, and re-rolling (with 97% survival) gives $0.97 \cdot EU(a)$:

$$EU(a) = \frac{1}{100} \sum_{x=a+1}^{100} \ln(1 + x) + \frac{a}{100} \cdot 0.97 \cdot EU(a)$$

Isolating $EU(a)$:

$$EU(a)\left(1 - \frac{0.97a}{100}\right) = \frac{1}{100}\sum_{x=a+1}^{100} \ln(1+x)$$

$$EU(a) = \frac{\sum_{x=a+1}^{100} \ln(1+x)}{100 - 0.97a} \tag{2}$$

Scanning equation (2) across all integer thresholds (same idea as the EV table in Part 2) gives a peak at $a = 57$. Plugging in:

$$\text{Numerator} = \sum_{x=58}^{100} \ln(1+x) = \ln(59) + \ln(60) + \cdots + \ln(101) = 187.8982$$

$$\text{Denominator} = 100 - 0.97 \times 57 = 100 - 55.29 = 44.71$$

$$EU_{\max} = \frac{187.8982}{44.71} \approx \boxed{4.2026}$$

Converting to the Certainty Equivalent. $EU_{\max} = 4.2026$ is a utility number, not a dollar amount as we cannot directly compare it to \$75. To bring it back to dollars, we invert the utility function. Since $U(x) = \ln(1+x)$, the inverse is $x = e^U - 1$:

$$CE_{\max} = e^{EU_{\max}} - 1 = e^{4.2026} - 1 = 66.86 - 1 \approx \boxed{\$65.86}$$

This is the guaranteed cash amount a risk-averse player values equally to the full game.

Now re-evaluating the three choices in dollar-equivalent terms:

Choice A: \$75.00

Choice B:

$$0.03 \times 0 + 0.97 \times CE_{\max} = 0.97 \times 65.86 \approx \$63.88$$

Choice C:

$$0.03 \times 75 + 0.97 \times CE_{\max} - 2.00 = 2.25 + 63.88 - 2.00 \approx \$64.13$$

The ranking flips:

$$\text{Choice A } (\$75.00) > \text{Choice C } (\$64.13) > \text{Choice B } (\$63.88)$$

A risk-averse agent stops and takes the \$75. The psychological discount applied to a fresh roll i.e. from \$80.90 down to \$65.86 and it is large enough that the certain cash in hand wins out. Insurance still beats going naked, but not by enough to justify re-rolling at all.

Final answer: Under risk-neutrality, buy insurance and re-roll (Choice C). Under risk-aversion, stop now (Choice A). The \$0.25 insurance edge exists in both cases, but a risk-averse player won't re-roll regardless of whether the insurance is fairly priced or not.